

SANA FESSUH

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EDUCATION

International Baccalaureate scholar

The University of Chicago, Chicago, IL

Bachelor of Science in Data Science

Minor in Media Arts and Design

Relevant Coursework: Data Modeling, Machine Learning, Applications of Data Analysis, Data Visualization, Data Cleaning, Data Mining & Analytics, Database Systems, Data Engineering, Spatial Clustering, Linear Algebra, Probability, Calculus

PROJECTS

Data Science Clinic – UChicago

November 2024 - June 2025

Student Consultant

Project 1: Power–Danger Narratives in African American YA Literature

- Designed an NLP pipeline to clean, tokenize, and score 56 novels using the Power–Danger framework.
- Applied NLP techniques and statistical analysis to detect narrative bias across African American vs. white-authored YA literature.
- Developed time-series and distribution visualizations (using empirical mode decomposition) to track shifts in power and danger word usage.

Project 2: Landslide Risk Prediction – Gitwe Kadhua Corridor

- Processed provided geospatial data (elevation, slope, soil type, precipitation) using QGIS to prepare features for modeling.
- Engineered features and implemented machine learning models (Random Forest, Regression, Neural Network) to predict landslide risk.
- Evaluated models across multiple performance metrics (accuracy, precision/recall, ROC-AUC) to determine the best-performing approach.

EXPERIENCE

IPPH COMPASS, UChicago

November 2024 - June 2025

Student Research Assistant

- Performed cleaning, organization, and matching of patient-level data from the Chicago Multiethnic Prevention and Surveillance Study to support health disparity analysis.
- Utilized the EPIC electronic health record system to access and match MRN data, ensuring accurate linkage of patient records for health disparity research.

Embodying Racism Lab, UChicago

November 2024 - Present

Research Assistant

- Collected, cleaned, and structured unformatted health, administrative, and policy data from sources including the Census and Current Population Survey to support research on maternal and child health disparities.
- Maintained data analysis workflows, codebooks, and repositories to ensure reproducibility across projects.
- Performed descriptive and preliminary statistical analyses to identify patterns and inform research findings.
- Prepared tables for manuscripts and policy reports in collaboration with cross-functional research teams.

SKILLS

Programming & Query Languages: Python, R, SQL

Data Visualization & Geospatial Tools: Dash, PowerBI (basic), Matplotlib, Seaborn, Plotly, Alteryx, QGIS, Geoda

Tools & Platforms: Jupyter Notebooks, VS Code, Excel (pivot tables, VLOOKUPS, validation), Git, GitHub, AWS, Google Cloud

Machine Learning & AI: PyTorch, Facial Expression Recognition

Web & Creative Coding: HTML, Java(basic), Flask, Django, p5.js, Sonic Pi, NodeBox, Field, Turtle, pygame

Languages: Amharic (Fluent)

Portfolio: <https://shfessuh.github.io/>